

2026考研线性代数基础班

主讲：晓千老师（满分教练）



主讲老师简介

姜晓干

中国人民大学博士

高教社考研数学系列图书

主编、主讲

时代 $3+1+1=32$

基础 { 行列式
矩阵
向量

☆ { 方程组
特征值
二次型

第一章 行列式

$$|A|=2, |A^{-1}|=\frac{1}{2}, |A^*|=2^{n-1}$$

√ 概念
重要
展开
公式
Cramer



专题一 行列式的概念

定义

性质 (5条)

逆序数的定义 在 $1, 2, \dots, n$ 的一个全排列中, 若两个数的前后次序与标准次序不同, 则称这两个数逆序. 一个排列中所有逆序的总数称为逆序数, 记作 τ .

大的数在小的数之前

例: $\tau(54321) = 4 + 3 + 2 + 1 = 10$

行列式的定义 n 阶行列式定义为

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} a_{1j_1} a_{2j_2} \cdots a_{nj_n}$$

$n \cdot (n-1) \cdots 1 = n!$ $j_1 \neq j_2 \neq \cdots j_n$

即 $n!$ 项不同行不同列元素乘积的代数数和。

【评注】由行列式的定义知 2 阶、3 阶行列式满足对角线法则，即

主-副

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc, \quad \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_3 b_2 c_1 - a_2 b_1 c_3 - a_1 b_3 c_2$$

pr: $= (1) ad - (2) bc = ad - bc$
 例如：爱的行列式

2阶 $3! = 6$ 阶 8阶 $4! = 24$ 阶

$$\begin{vmatrix} \text{我} & 0 & \text{生} \\ 0 & \text{有} & 0 \\ \text{你} & 0 & \text{幸} \end{vmatrix} = \text{我有幸一生有你}, \quad \begin{vmatrix} 5 & 2 & 1 \\ 1 & 2 & 5 \\ 34 & 1 & 34 \end{vmatrix} = 520$$

【例 1.1】(2021, 数二、三) 多项式 $f(x) =$

$$\begin{vmatrix} x & x & 1 & \underline{2x} \\ 1 & \underline{x} & 2 & -1 \\ 2 & 1 & \underline{x} & 1 \\ \underline{2} & -1 & 1 & x \end{vmatrix}$$

中 x^3 项的系数为_____.

【详解】

含有 x^3 的项为 $\overset{\tau(2134)}{+1} x^3 + \overset{\tau(4231)}{+1} 4x^3 = -x^3 - 4x^3$
 $= -5x^3$

故 x^3 项系数为 -5.

行列式的性质

(1) 行列互换, 行列式的值不变;

(2) 两行 (或列) 互换, 行列式变号;

(3) 提公因子;

例: 以二阶为例 $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} c & d \\ a & b \end{vmatrix} = bc - ad = -(ad - bc) = - \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = kad - kbc = k(ad - bc) = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

(4) 拆行（或列）分配，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & (a_{1j} + a'_{1j}) & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & (a_{2j} + a'_{2j}) & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & (a_{nj} + a'_{nj}) & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a'_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a'_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a'_{nj} & \cdots & a_{nn} \end{vmatrix}$$

(5) 一行（或列）乘 k 加到另一行（或列），行列式的值不变.

推论 两行（或列）成比例，行列式为零.

【例 1.2】

$$\begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} = \underline{\hspace{2cm}}.$$

【详解】

$$\begin{aligned} \text{解} \quad \begin{vmatrix} a^2 & 2a+1 & 4a+4 & 6a+9 \\ b^2 & 2b+1 & 4b+4 & 6b+9 \\ c^2 & 2c+1 & 4c+4 & 6c+9 \\ d^2 & 2d+1 & 4d+4 & 6d+9 \end{vmatrix} &= \begin{vmatrix} 2 & 6 \\ 2 & 6 \\ 2 & 6 \\ 2 & 6 \end{vmatrix} = 0 \end{aligned}$$

专题二 重要行列式 (F14)

1. 上（或下）三角、主对角行列式

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} = a_{11}a_{22}\cdots a_{nn}$$

$\begin{matrix} \text{pr:} \\ \text{=} \end{matrix} \overset{Z(1,2 \dots n)}{=} a_{11} \cdots a_{nn}$
 $= a_{11} \cdots a_{nn}$

2. 关于副对角线的上（或下）三角、副对角行列式

$$\begin{vmatrix} a_{11} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & \cdots & a_{2,n-1} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,n-1} & a_{nn} \end{vmatrix} = \begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix} = (-1)^{\boxed{\frac{n(n-1)}{2}}} a_{1n} a_{2,n-1} \cdots a_{n1}$$

$$n-1 + n-2 + \cdots + 1$$

$$\begin{aligned}
 & \text{pr: } = (-1)^{z(n, n-1, \dots, 1)} a_{1n} a_{2,n-1} \cdots a_{n1} \\
 & = (-1)^{\frac{n(n-1)}{2}} a_{1n} \cdots a_{n1}
 \end{aligned}$$

3. n 阶 ab 型行列式

$$\begin{vmatrix} a & b & \cdots & b \\ b & a & \cdots & b \\ \vdots & \vdots & \ddots & \vdots \\ b & b & \cdots & a \end{vmatrix} = [a + (n-1)b](a-b)^{n-1}$$

证：每行元素之和相等，每列加到第1列

$$\text{原式} = [a + (n-1)b] \begin{vmatrix} b & \cdots & b \\ a & \cdots & b \\ \vdots & \ddots & \vdots \\ b & \cdots & a \end{vmatrix}$$

$$= (a + (n-1)b) \begin{vmatrix} 1 & b & \cdots & b \\ 0 & a-b & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a-b \end{vmatrix}$$

$$= (a + (n-1)b) (a-b)^{n-1}$$

补例（1997，数四）设 n 阶矩阵

$$A = \begin{pmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{pmatrix}$$

则 $|A| = \frac{f(1)(n-1)}{n}$.

4. 拉普拉斯展开式

(分块矩阵的行列式)

设 A 为 m 阶矩阵, B 为 n 阶矩阵, 则 一个字母代表一个小块, 看成一块

Laplace 1749

$$\begin{vmatrix} A & C \\ O & B \end{vmatrix} = \begin{vmatrix} A & O \\ C & B \end{vmatrix} = \begin{vmatrix} A & O \\ O & B \end{vmatrix} = |A||B|$$

$$\begin{vmatrix} C & A \\ B & O \end{vmatrix} = \begin{vmatrix} O & A \\ B & C \end{vmatrix} = \begin{vmatrix} O & A \\ B & O \end{vmatrix} = (-1)^m |A||B|$$

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$$\text{Pr: } \begin{vmatrix} 0 & A \\ B & 0 \end{vmatrix} = \begin{vmatrix} \begin{matrix} \text{O} \\ \vdots \\ \text{O} \end{matrix} & \begin{matrix} a_{11} & \dots & a_{1m} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mm} \end{matrix} \\ \begin{matrix} b_{11} & \dots & b_{1n} \\ \vdots & & \vdots \\ b_{m1} & \dots & b_{mn} \end{matrix} & \begin{matrix} \text{O} \\ \vdots \\ \text{O} \end{matrix} \end{vmatrix} \xrightarrow{+} \begin{vmatrix} \text{O} & A \\ 0 & B \end{vmatrix} \\
 = \text{Pr}^{mn} (A|B)$$

The diagram illustrates the expansion of a block determinant. The first matrix is a 2×2 block matrix with blocks 0 , A , B , and 0 . This is equated to a larger matrix where the top-left block is a column of zeros, the top-right block is matrix A (with elements a_{ij}), the bottom-left block is matrix B (with elements b_{ij}), and the bottom-right block is a column of zeros. The matrix is partitioned by vertical lines. The top-right block A is circled in red, and the bottom-right block (a column of zeros) is also circled in red. Purple arrows point to the top-right and bottom-right blocks. The matrix is then transformed into a block upper triangular form, where the bottom-left block B is zeroed out, resulting in a block diagonal matrix with blocks A and B . The final expression is $\text{Pr}^{mn} (A|B)$.

【例 1.3】(2014, 数一、二、三)

$$\begin{vmatrix} 0 & a & b & 0 \\ a & 0 & 0 & b \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix}$$

【 】 $\overset{\text{这是错误}}{=} |A||P||B||C|$

(A) $(ad - bc)^2$

(B) $-(ad - bc)^2$

(C) $a^2d^2 - b^2c^2$

(D) $b^2c^2 - a^2d^2$

【详解】

$$\begin{vmatrix} a & b & 0 & 0 \\ 0 & 0 & a & b \\ c & d & 0 & 0 \\ 0 & 0 & c & d \end{vmatrix} = - \begin{vmatrix} a & b & 0 \\ c & d & a \\ 0 & c & b \end{vmatrix} = -(ad - bc)^2$$

补例 (1996, 数一)

$$\begin{vmatrix} a_1 & 0 & 0 & b_1 \\ 0 & a_2 & b_2 & 0 \\ 0 & b_3 & a_3 & 0 \\ b_4 & 0 & 0 & a_4 \end{vmatrix} = \mathbf{【 \quad \quad \quad 】}$$

$$= \begin{vmatrix} a_1 & b_1 & 0 & 0 \\ 0 & 0 & a_2 & b_2 \\ 0 & 0 & b_3 & a_3 \\ b_4 & a_4 & 0 & 0 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & 0 \\ b_4 & a_4 & 0 \end{vmatrix} \begin{vmatrix} a_2 & b_2 \\ b_3 & a_3 \end{vmatrix} = (a_1 a_4 - b_1 b_4)(a_2 a_3 - b_2 b_3)$$

(A) $a_1 a_2 a_3 a_4 - b_1 b_2 b_3 b_4$

(B) $a_1 a_2 a_3 a_4 + b_1 b_2 b_3 b_4$

(C) $(a_1 a_2 - b_1 b_2)(a_3 a_4 - b_3 b_4)$

(D) $(a_2 a_3 - b_2 b_3)(a_1 a_4 - b_1 b_4)$

【详解】

△ Laplace 的推广

【例 1.4】 设 A 为 m 阶矩阵, B 为 n 阶矩阵, C 为 l 阶矩阵, $D = \begin{pmatrix} O & O & A \\ O & B & O \\ C & O & O \end{pmatrix}$. 若 $|A| = a$, $|B| = b$,

$|C| = c$, 则 $|D| =$ _____.

【详解】

$$\begin{aligned} |D| &= (-1)^{m(m+l)} |A| \begin{vmatrix} O & B \\ C & O \end{vmatrix} \\ &= (-1)^{mn+ml} |A| \cdot (-1)^{nl} |B| |C| \\ &= (-1)^{mn+ml+nl} abc \end{aligned}$$

5. 范德蒙行列式

$$D_n = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix} = \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

pr: 归纳/内法 当 $n=2$ 时 $\begin{vmatrix} 1 & 1 \\ x_1 & x_2 \end{vmatrix} = x_2 - x_1$

【例 1.5】
$$\begin{vmatrix} 6 & 7 & 8 & 9 \\ 4^2 & 3^2 & 2^2 & 1 \\ 4^3 & 3^3 & 2^3 & 1 \\ 4 & 3 & 2 & 1 \end{vmatrix} = \text{—————}.$$

【详解】
$$\text{行列式} = 10 \begin{vmatrix} 1 & 1 & 1 & 1 \\ 4 & 3 & 2 & 1 \\ 4^2 & 3^2 & 2^2 & 1 \\ 4^3 & 3^3 & 2^3 & 1 \end{vmatrix} = 10 \times 12 = 120$$

【例 1.6】 设 a_1, a_2, a_3, a_4 为非零常数, 则

$$\begin{vmatrix} a_1^3 & a_1^2 b_1 & a_1 b_1^2 & b_1^3 \\ a_2^3 & a_2^2 b_2 & a_2 b_2^2 & b_2^3 \\ a_3^3 & a_3^2 b_3 & a_3 b_3^2 & b_3^3 \\ a_4^3 & a_4^2 b_4 & a_4 b_4^2 & b_4^3 \end{vmatrix} = \underline{\hspace{2cm}}.$$

【详解】

第 i 行提 $a_i^3, i=1, \dots, 4$ 得

$$\text{原式} = a_1^3 \cdots a_4^3 \begin{vmatrix} 1 & \frac{b_1}{a_1} & (\frac{b_1}{a_1})^2 & (\frac{b_1}{a_1})^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \frac{b_4}{a_4} & (\frac{b_4}{a_4})^2 & (\frac{b_4}{a_4})^3 \end{vmatrix} = a_1^3 \cdots a_4^3 \prod_{1 \leq i < j \leq 4} \left(\frac{b_j}{a_j} - \frac{b_i}{a_i} \right)$$